



ISBT 128 STANDARD

Chain of Identity (Col) Identifier

DRAFT

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1 Introduction

1.1 Purpose

The purpose of this document is to provide:

- specifications for the structure of the Chain of Identity (Col) Identifier
- information on how to obtain and update a Facility Identification Number (FIN)
- rules on the use of the Col Identifier

Throughout this document where the word “shall” is used, it represents a requirement; where the word “should” is used, it represents a recommendation; and where the word “may” is used, it represents an option.

1.2 Scope

This standard specifies the rules for the structure and use of the ISBT 128 Col Identifier. It also provides information on the associated Facility Identification Number (FIN).

1.3 Intended Audience

The intended audience of this document is cellular therapy collection and processing facilities, clinical trials sponsors, manufacturers of cellular therapy products, software developers, laboratory staff, and staff responsible for the clinical application of products carrying a Col identifier.

1.4 Normative Reference

[*ISBT 128 Standard Technical Specification \(ST-001\)*](#)

[*ISBT 128 Standard Labeling of Collection Products for Cellular Therapy Manufacturing \(ST-018\)*](#)

ISO/IEC 7064:2003(E): Information technology—Security techniques—Check character systems

ISO/IEC 15417: 2007(E): Information technology—Automatic Identification and data capture techniques—Code 128 bar code symbology specification

ISO/IEC 16022:2006(E): Information technology—International symbology specification—Data Matrix (and correction ISO/IEC 16022:2006/Cor 1:2008)

1.5 Other References

[*Implementation Guide: ISBT 128 Facility Identification Number \(IG-034\)*](#)

ICCBBA Website (www.isbt128.org)

1.6 Background

The ISBT 128 Donation Identification Number (DIN) provides a globally unique identifier that can be used to identify all products derived from a single donation. The DIN is also suitable for identifying pools of donations where the pooling process is performed at an ICCBBA licensed facility.

However, there are circumstances in the collection and processing of cellular therapy products for further manufacture where more than one donation may need to be collected to deliver a specific therapy event. To ensure that the individual DINs of the collected donations can be linked, and to provide a single identifier for downstream products, a new identifier, the CoI Identifier is required. The CoI Identifier may be allocated either before, or at the time of, collection of the first donation.

The CoI Identifier will not replace the DIN on the collection products but will be an additional identifier that is the same on all donations associated with a single patient therapy event.

The CoI Identifier can be utilized on both ISBT 128 Standard labels and non-ISBT 128 labels.

The responsibility for allocating the CoI Identifier may lie with the collecting facility, the clinical trials sponsor, or manufacturers of cellular therapy products. This specification supports these approaches.

2 Format and Purpose of the FIN in the Col Identifier

ICCBBA assigns Facility Identification Numbers (FINs) to facilities that are licensed to use ISBT 128. The FIN is a five-character alphanumeric code that can be used in a variety of ways to ensure the uniqueness of an identification number and thus is essential to traceability. In the context of the Col Identifier, the FIN identifies the organization that issued the Col Identifier.

The FIN shall be encoded and interpreted by reference to the ICCBBA Facility Identification Number Database published and maintained by ICCBBA on the ICCBBA Website.

Any organization licensed with ICCBBA may use any FIN assigned to it within a Col Identifier. Any organization not currently licensed with ICCBBA can apply for registration and will be allocated a FIN.

The FIN in the Col Identifier is present as a means of ensuring global uniqueness across multiple organizations. However, it is not intended to be parsed as a data item in its own right to identify the organization with which the product is associated. Where it is necessary to transmit the identity of this organization the FIN should be used in a data field specifically designed for the purpose.

A range of FINs has been reserved for validation testing. This range is A9990 through A9999. Facilities should use FINs within this range in a COI when performing validation testing. This range may also be used for example labels when the use of an actual FIN is not recommended (e.g., a facility wanting to show an example label for educational purposes).

Further information on FIN and the maintenance of the FIN Database is available in [Implementation Guide: ISBT 128 Facility Identification Number \(IG-034\)](#)

3 Col Identifier Issuing Organizations

Col Identifier issuing organizations shall be registered and licensed with ICCBBA.

Col Identifier issuing organizations shall ensure that they maintain licensed status and inform ICCBBA of any necessary updates to their organization's registration information.

Col Identifier issuing organizations shall only issue Col Identifiers that contain a FIN assigned to their organization.

Col Identifier issuing organizations shall have systems in place to control the issue of Col Identifiers in a manner that ensures uniqueness.

4 Col Identifier Allocation and Presentation Rules

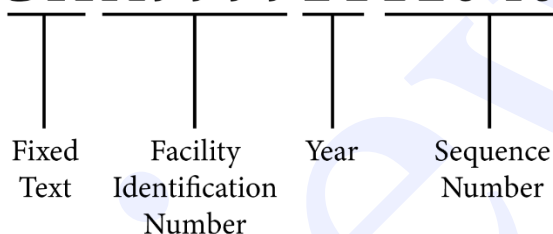
4.1 Allocating a Col Identifier

The Col Identifier is a 15-character identifier composed of four elements: the fixed two-character string “CH”, a five-character FIN; a two-digit year of issue; and a six-digit sequence number assigned by the Issuing Organization.

Figure 1 Col Identifier Example

Chain of Identity (Col) Identifier

CHA999921123456



An organization issuing a Col Identifier shall use a FIN assigned to the organization.

The year indicator ensures uniqueness over a 100-year period. This is a nominal year indicator and may overlap +/- one month of the year assigned.

Sequence numbers shall be controlled in such a manner that when combined with the fixed text, FIN, and year code they uniquely identify a single therapy event.

Once assigned a Col Identifier shall not be reassigned.

4.2 Eye-readable presentation of the Col Identifier

Eye-readable versions of the Col Identifier shall be accompanied by a boxed manual entry checksum (see section 4.3).

When printed in an eye-readable format, the Col Identifier may be divided into blocks using spaces to assist manual transcription.

Spacing between the blocks shall be sufficient to ensure the blocks are clearly separated.

The Col Identifier shall be printed in a font that allows differentiation between similar letters and digits (i.e., 0 and O, 1 and I)

CH A9999 21 123456 V

Checksums shall be calculated as indicated in section 4.3 below.

4.3 Calculating the Checksum

The checksum is based on the ISO/IEC 7064 Mod 37-2 algorithm. This section shows how the checksum shall be calculated for any given Col Identifier. The calculation is based on the fifteen data characters of the Col Identifier.

The steps in the process are as follows:

1. For each character in the 15-character string, determine its check value as required by ISO 7064 from Table 1 (e.g., character F has value 15);
2. For each character in the 15-character string, determine its weighted check value by multiplying the check value from Table 1 by the n^{th} power of 2 where n is the position of the character from the right-hand end of the string;
3. Sum the weighted check values from step 2;
4. Find the modulus 37 value of the sum from step 3;
5. Subtract the value obtained in step 4 from 38;
6. Find the modulus 37 value of the result of step 5;
7. Using the value in step 6, determine the check character by referring to Table 1 (this time read the character from the value). This is the modulo 37-2 checksum character.

Table 1 Character to ISO/IEC 7064 Check Values [RT035]

Character	0	1	2	3	4	5	6	7	8	9	A	B	C
Value	0	1	2	3	4	5	6	7	8	9	10	11	12
Character	D	E	F	G	H	I	J	K	L	M	N	O	P
Value	13	14	15	16	17	18	19	20	21	22	23	24	25
Character	Q	R	S	T	U	V	W	X	Y	Z	*		
Value	26	27	28	29	30	31	32	33	34	35	36		

Example of Checksum Calculation

Col Identifier is **CHA999921123456**

Character in the string	STEP 1 ISO 7064 check value (a)	n Position of the character from the right	2^n (b)	STEP 2 Weighted check value (a x b)
C	12	15	32768	393216
H	17	14	16384	278528
A	10	13	8192	81920
9	9	12	4096	36864
9	9	11	2048	18432
9	9	10	1024	9216
9	9	9	512	4608
2	2	8	256	512
1	1	7	128	128
1	1	6	64	64
2	2	5	32	64
3	3	4	16	48
4	4	3	8	32
5	5	2	4	20
6	6	1	2	12

Step 3: sum of last column = 823664

Step 4: modulo 37 of 823664 = 7

Step 5: $38 - 7 = 31$

Step 6: modulus 37 of 31 = 31

Thus, the mod 37-2 checksum is V.

CH A9999 21 123456 V

Computer programs for calculating the checksum using ISO 7064 are described in Appendix A.3 of [ISBT 128 Standard Technical Specification \(ST-001\)](#)

A checksum calculator (Quick K Calculator) is located in the lookup tools section on the ICCBBA Website (www.isbt128.org)

4.4 Verifying the Col Identifier Checksum

The integrity of a manually transcribed Col Identifier should be verified each time a Col Identifier is received by a computer system through manual entry by performing a checksum calculation as indicated in section 4.3 above.

The calculated value should be checked against the character entered by the user. If the verification fails an error should be indicated and the entry repeated.

4.5 Electronic encoding of the Col Identifier

When the Col Identifier is represented in automatic identification and data capture (AIDC) solutions, it shall be encoded as Data Structure 040 in the ISBT 128 Standard.

Purpose: Data Structure 040 shall specify an ISBT 128 Col Identifier.

Structure: &/CHappppyynnnnnn

Element	Length	Type
&/	2	data identifiers
CH	2	literal "CH"
α	1	Alphanumeric {A–N; P–Z; 1–9}
pppp	4	First two characters alphanumeric {A–N; P–Z; 0–9}; second two characters numeric {0–9}. Current usage is numeric for all four characters. Alpha characters may be introduced into positions 1 and 2 in the future (e.g., if α = A and pppp = BC12, the appppp will be ABC12).
yy	2	numeric {0-9}
nnnnnn	6	numeric {0-9}

The data content string shall be 15 characters and shall be encoded and interpreted as follows:

CH	The literal string "CH"
αpppp	The five-character Facility Identification Number of the Issuing Organization
yy	Two-digit year indicator. (Nominal year of issue)
nnnnnn	Six-digit sequence number assigned by the Issuing Organization

If the Col Identifier is to be represented in a linear bar code, Code 128 shall be used and comply with ISO/IEC 15417: 2007: Information technology—Automatic identification and data capture techniques—Code 128 bar code symbology specification.

Figure 2 Col Identifier linear bar code example



If the Col Identifier is to be represented in a 2-D symbol, Data Matrix shall be used and comply with ISO/IEC 16022:2006: Information technology— Automatic identification and data capture techniques—Data Matrix bar code symbology specification (including the corrections ISO/IEC 16022:2006/Cor 1:2008 and ISO/IEC 16022:2006/Cor 2:2011).

The Col Identifier may be combined with additional information in the 2-D symbol as part of an ISBT 128 Compound Message. Additional information about the Compound Message data structure and requirements for the use of Code 128 and Data Matrix is found in the [ISBT 128 Standard Technical Specification \(ST-001\)](#).

4.6 Col Identifier in Electronic Messages

The Col Identifier can be represented in electronic messages using the data element ChainOfIdentityIdentifier with a unique resource identifier (URI) of <https://www.isbt128.org/uri/ChainOfIdentityIdentifier>

The data value for this data element shall be the 15 data characters of the Col.

XML Example:

```
<ChainOfIdentityIdentifier Identifier=https://www.isbt128.org/uri/ChainOfIdentityIdentifier
value="CHA999921123456"/>
```

4.7 Label examples

The following labels illustrate how the Col Identifier can be used to link two separate collections intended for the same therapy event. The label examples in both Figure 3 and Figure 4 are apheresis collection products for cellular therapy manufacturing as described in [ISBT 128 Standard Labeling of Collection Products for Cellular Therapy Manufacturing \(ST-018\)](#). In these examples, the Col Identifier encodes the FIN of the cell therapy manufacturer, while the DIN encodes the FIN of the collection facility. Both the Col Identifier and the DIN are encoded in the 2-D symbol along with the Product Code and Expiration Date. These labels are provided as examples only. National regulations and appropriate standards must be consulted to ensure full compliance with requirements.

Figure 3 Label example – Col Identifier provided by cell therapy manufacturer

		For Clinical Trial Use Only FOR AUTOLOGOUS USE ONLY	
A9996 22 123458 8 1		Intended Recipient: Recipient ID: XXN127654 DOE, Charlie Alex Patient DOB: 1999-06-01	
Collection Center City, State, Country, Postal Code		Expiration Date/Time: 2022-05-17 13:40 EST (2022-05-17 18:40 UTC)	
Collection Date and Time 2022-05-14 13:40		COI: CH A9999 22 123456 E	
Do Not Irradiate		Protocol: NCT99999999	
		Collection Center Site No: 47 Autologous Apheresis for Further Use in Manufacturing of XXXXXX Drug Product	
S1303100			
MNC, APHERESIS For Further Processing		Sponsor Info/Logo Area	
Total Volume ____ mL containing approx ____ mL Citrate			
Store at 1 to 10 C			

Figure 4 Label example – Col Identifier provided by cell therapy manufacturer, subsequent collection for the same therapy event as Figure 3

	For Clinical Trial Use Only FOR AUTOLOGOUS USE ONLY
A9996 22 123471 8 7 Collection Center City, State, Country, Postal Code	Intended Recipient: Recipient ID: XXN127654 DOE, Charlie Alex Patient DOB: 1999-06-01
Collection Date and Time 2022-05-20 13:40	Expiration Date/Time: 2022-05-23 13:40 EST (2022-05-23 18:40 UTC)
Do Not Irradiate	COI: CH A9999 22 123456 E Protocol: NCT99999999
 S1303100	Collection Center Site No: 47 Autologous Apheresis for Further Use in Manufacturing of XXXXXX Drug Product
MNC, APHERESIS For Further Processing	
Total Volume ____ mL containing approx ____ mL Citrate	Sponsor Info/Logo Area
Store at 1 to 10 C	